# Design a Robot with Balancing Ability and Self-controllability

Zhihao Lin*[1] and Zongjun Yang†[1]

[1]*Research School of Physics, Australian National University, Canberra ACT, 2601, Australia*

## 1. EXECUTIVE SUMMARY / ABSTRACT

This report presents a comprehensive design review of an autonomous two-wheeled self-balancing robot, including system architecture, physical modeling, and current prototype progress. The primary objective is to develop a robot capable of maintaining dynamic balance, resisting external perturbations, and executing controlled motion.

The proposed system architecture is driven by an Arduino UNO with a high-frequency Proportional-Integral-Derivative (PID) feedback control. Real-time spatial states sensing is acquired via an $I^2C$ MPU-6050 IMU. Actuation is delivered by two geared DC motors controlled by an L298N motor driver. Furthermore, the 3D-printed physical chassis was modeled using Lagrangian mechanics, intentionally increasing the height of mass center to lower the falling speed and provide the processor a temporal window for actuation.

As of Week 6, the project is progressing ahead of schedule. The physical prototype has been fully assembled, and the core balancing functionality has been successfully demonstrated. The robot can maintain steady-state equilibrium indefinitely with minimal drift ($< 0.15m$ in 1s). While the current hardware configuration recovers from disturbances up to only 13° (less than the 15° target), the direction of mechanical and control improvements have been identified to resolve this during the next phase.

Moving forward, the team has transitioned to a concurrent engineering approach to develop advanced mobility and adaption skills in parallel. A revised timeline is used to map the final integration and comprehensive testing phases across Weeks 7-12.

## 2. INTRODUCTION

### 2.1. Background

There are increasing applications that need robots to participate. For example, complicated transportation in lab environments [**application_lab_transportation_liu_2013**], security support with interaction ability [**application_security_lopez_2017**], and educational communication [**application_social_tony_2018**]. With the revolution of machine learning and artificial intelligence, robots are becoming much more intelligent [**ai_soori_2023**, **mobile_robot_raj_2022**], which will be applied to wider field.

Fundamentally, a robot requires several characteristics including: it must be produced by manufacture, it must be able to move itself or other objects, and it must be able to respond to its environment [**fundamental_todd_2012**]. This indicates the essential parts of a robot: physical bodies, action drivers, sensors, and processors (controllers). Mobile robot is a typical case of both robot definition and robot applications [**mobile_robot_tzafestas_2018**, **mobile_robot_raj_2022**].

To implement robots to everyday applications, we need to start from a prototype, and upgrade it step by step. The objective of this project is to design, fabricate, and program an autonomous robot with excellent balancing performance and high controllability.

### 2.2. Requirements Review

The primary objective of this project is to build an autonomous self-balancing robot which is able to move to or keep itself around a designated position. To achieve this, the listed specific requirements are as follows:

- Top-level requirements:
  - The robot must maintain its balanced position without any falling cases.
  - If the robot is perturbed externally (e.g., a deliberated push, or an obstacle collided), it must be able to return balanced within $2s$. The perturbation is defined as changing its pitch angle up to 15°.
  - Without perturbation, the robot must balance itself in a small range. It must not move more than $0.5m$ in $1s$.
- Electronic components:
  - The robot must be able to sense its angle or velocity at $> 100Hz$. The resolution of angle must be $< 0.5°$.
  - The robot must take action within $0.1s$ after a controlling signal is sent.
  - The electronics must be stable during its runtime.
  - After turning on, the robot must work entirely independently, including power supply, signal processing, and controlling.
- Physical structure:
  - The structure must be stable enough that the position of mass center doesn't change.
  - The chassis must be sturdy, which can resist damage when the robot falls in some cases.
- Advanced requirements:
  - The robot can be able to fully control its state, including facing direction and moving speed.
  - The robot can follow a designated path without obvious drift.
  - The robot can dynamically adapt its response to varying operational environments: in peaceful environments, being slightly perturbed, being strongly perturbed, commanded to move to and keep around a position.

## 3. TECHNICAL DETAILS AND ANALYSIS

### 3.1. System Architecture

Fig. 1 shows the flowchart of the basic balancing idea of the robot. Generally, it shows the realization of the basic balancing skills. Any other advanced skills can be add to these parts. Reading multiple types of states (e.g., pitch angle, yaw angle, and x-acceleration) can be add in the "Read State" process. "Check States" decision can be used to assess the situation of the robot. "Feedback Controller" process can handle different controllers, such as PID controller. "Take Action" process controls the states of the robot.

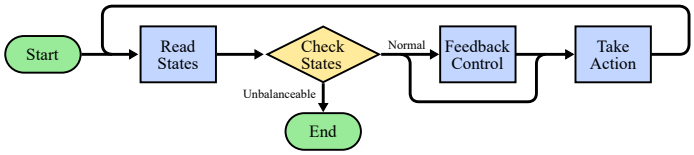



**Figure 1.** Flowchart of basic balancing logic. After turning on, the robot keeps reading its state, such as the pitch angle or the angular velocity of y-axis. If it's not strongly unbalanced, feedback controllers are used to generate an output from the state. The output is used to control the robot, such as moving forward or turn left. The feedback loop can balance the robot theoretically.

We started from developing the basic balancing skill. When this first stage was completed successfully, we would try to realize the advanced requirements by upgrading the existed processes or decisions instead of rebuild a new structure. In this way, the development would be

*u8175491@anu.edu.au
†u8463895@anu.edu.au

controllable and the progress would be clear. However, a new structure can be required if we want to develop a more complex logic.

## 3.2. Electronic Components

To achieve the main goal, the electronic components should contain these necessary parts: a state sensor, a central processor, an action driver, and power supply.

### 3.2.1. The state sensor

We use MPU-6050 inertial measurement unit (IMU) as the state sensor. It consists of a triple-axis gyroscope and a triple-axis accelerometer. Using the angular velocities and linear accelerations, we can get or calculate most of the important states of robot. The key parameters of MPU-6050 are listed in Table 1:

| Parameter | Value | Unit |
|---|---|---|
| Gyroscope Full-scale Range | $\pm 250$ | $^\circ/s$ |
| Gyroscope Sensitivity | 131 | $LSB/(^\circ/s)$ |
| Gyroscope Noise Spectral Density | 0.005 | $^\circ/s/\sqrt{Hz}$ |
| Gyroscope Output Rate | 8000 | $Hz$ |
| Accelerometer Full-scale Range | $\pm 2$ | $g$ |
| Accelerometer Sensitivity | 16384 | $LSB/g$ |
| Accelerometer Noise Spectral Density | 400 | $\mu g/\sqrt{Hz}$ |
| Accelerometer Output Rate | 1000 | $Hz$ |
| Supported Communication | $I^2C$ | |
| Communication Speed | 400 | $kHz$ |
| Clocking Speed | $32.768k \sim 19.2M$ | $Hz$ |
| Power Input Voltage | $2.375 \sim 3.46$ | $V$ |

**Table 1.** The key parameters of MPU-6050 inertial measurement unit [**mpu6050_datasheet**]

The best resolution of angular velocity can reach $7.6\times10^{-3}$ $^\circ/s$, and the best resolution of acceleration can reach $61 \times 10^{-6}$ $g$. Ignoring the noise, these resolutions are totally sufficient for the requirements. Both the sampling rate and communication speed satisfy our requirements. The primary limitation of this IMU is the inherent sensor noise or drift. Considering the noise, we can try to process the signal (such as low-pass filter or quaternion) to get high enough resolution.

### 3.2.2. The central processor

Arduino UNO is chosen as the processor. We can use it to get data from the IMU, process the data, and control the action driver. Its key parameters are listed in Table 2:

| Parameter | Value | Unit |
|---|---|---|
| # Digital I/O Pins | 14 | |
| # Analog Input Pins | 6 | |
| # PWM Pins | 6 | |
| Supported Communication | $I^2C, SPI, UART$ | |
| Clocking Speed | 16 | $MHz$ |
| Power Output Voltage | 3.3, 5 | $V$ |
| Power Input Voltage | $7 \sim 12$ | $V$ |

**Table 2.** The key parameters of Arduino UNO [**arduino_uno_datasheet**]

The number of digital, analog and PWM pins are enough for the wiring connecting the processor to the sensor and the action driver. The high processing speed is ideal for our control logic. Using $I^2C$ communication and $3V3$ output power, it can match the MPU-6050 very well. However, the power output voltage may not be enough for some action drivers. So a shared power supply may be required.

### 3.2.3. The action driver

We choose to use a L298N motor driver and two geared DC motors. The key parameters of the L298N motor driver are listed in Table 3:

| Parameter | Value | Unit |
|---|---|---|
| Digital Input Pins | $IN1 + IN2, IN3 + IN4$ | |
| PWM Input Pins | $ENA, ENB$ | |
| Logic Input Voltage | 5 | $V$ |
| Power Output Voltage | $3.2 \sim 40$ | $V$ |
| Power Input Voltage | $5 \sim 35$ | $V$ |

**Table 3.** The key parameters of L298N motor driver [**l298n_datasheet**]

The motor driver allows PWM input, which matches well with the Arduino UNO. The digital input pins can be used to control the direction of motors, while the PWM input pins can be used to control the speed. The required voltage is high as expected, so a good power supply is needed. However, by experiments, we found that the motors can keep still when the speed signal is small, which is caused by the internal static friction. Feed-forward controller can be added to solve this dead-band. Moreover, the two motors have a little asymmetry, which makes the robot yaw. Therefore, stepper motors can be required when we need to do precise control or measurement.

### 3.2.4. Power supply

We need to supply power to both the Arduino UNO and the motor driver. Therefore, two 800mAh 3.7V rechargeable Li-ion cells are used. For reusability, we use a breadboard to transfer the power from batteries to the two components instead of soldering the wires. The safe charged voltage of Li-ion is $2.8 \sim 4.2$ $V$. By testing, the batteries need to be recharged after working for about 10 hours.

### 3.2.5. Wiring structure

As mentioned above, the electronic components used are:

- MPU-6050 IMU
- Arduino UNO
- L298N motor driver
- Geared DC motors
- Li-ion Batteries
- Breadboard

The wiring method is shown in Fig. 2.

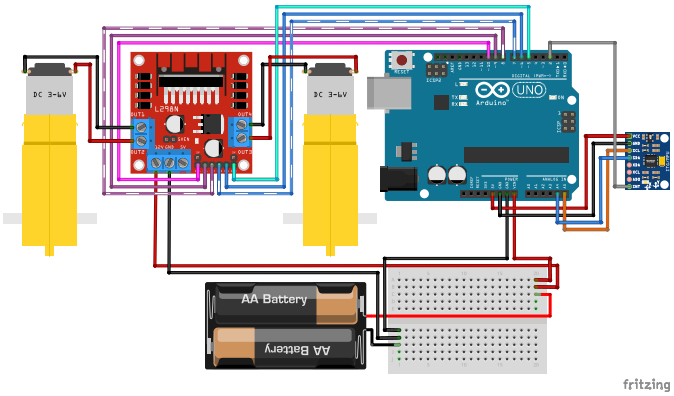



**Figure 2.** Wiring diagram of the electronic components. The battery and breadboard are used to power the Arduino UNO and L298N motor driver. UNO communicates with MPU-6050 inertial measurement unit (IMU) using I2C protocol, and sends analog signal to the motor driver using pulse width modulation (PWM). The motor driver controls 2 motors.

## 3.3. Mechanical Design

### 3.3.1. Physical model analysis

Fig. 3 shows the simplified physical model of the robot. We can use Lagrangian mechanics to analyze the relationship between $F$ and $\theta$, $x$. The system Lagrangian and Euler–Lagrange equations are given by:

$$L = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left[(\dot{x} + l\dot{\theta}\cos\theta)^2 + (l\dot{\theta}\sin\theta)^2\right] - mgl\cos\theta \quad (1)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = F \quad \Rightarrow \quad (M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = F \quad (2)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0 \quad \Rightarrow \quad \ddot{x}\cos\theta + l\ddot{\theta} - g\sin\theta = 0 \quad (3)$$

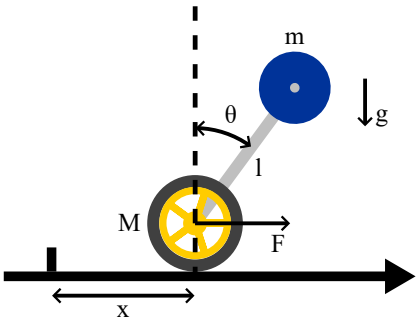



**Figure 3.** A simplified physical model of the robot. Constrain the movement in 1 dimension. THe model consists of a wheel with mass $M$, a body mass center $m$, and a light stick connecting these two with length $l$. The action of wheel is simplified as a force $F$. Assume the robot is at position $x$ and its pitch angle is $\theta$.

Usually, we care about the dynamic of $\theta$, which can be derived from Equation 2 and Equation 3:

$$\ddot{\theta} = \frac{(M+m)g\sin\theta - F\cos\theta}{(M + m\sin^2\theta)l} - \frac{m\dot{\theta}^2\sin 2\theta}{2(M + m\sin^2\theta)} \quad (4)$$

According to the model Fig. 3 and this Equation 4, we can analyze some cases:

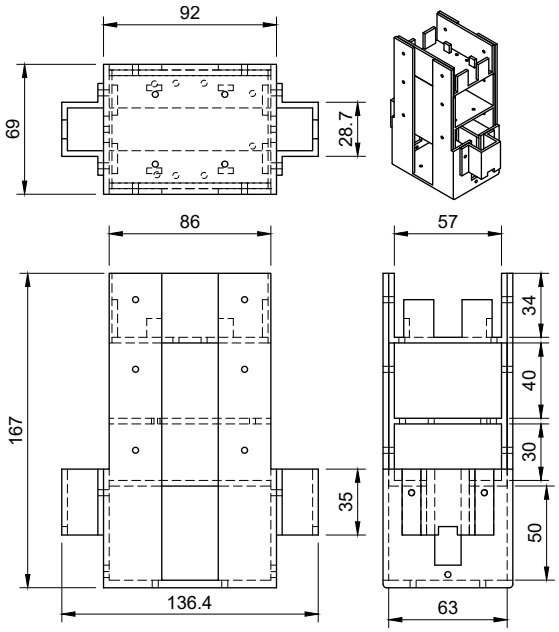
- The robot is perfectly at the balanced state, but a perturbation happens before the controller realizes it (i.e., $\theta \gtrapprox 0$, $F = \dot{\theta} = 0$). In this case, $\ddot{\theta} \approx \frac{(M+m)g\sin\theta}{lM}$. To maximize the controller's temporal response window before $\theta$ reaches an unbalanceable angle, we want $\ddot{\theta}$ to be as small as possible, which means we need to use small $m$ and/or large $l$.
- The robot is in a state of dynamic equilibrium (i.e., $\theta > 0$, $\ddot{\theta} \approx \dot{\theta} \approx 0$). To keep a constant $\theta$, it's necessary to make $F = (M + m)g\tan\theta$. However, because of the limit of motor power, $F$ can be maintained for only a short time before the large speed brings large resistance. So we should make $F > (M+m)g\tan\theta$ to bring back balance before the saturation of motors.
- The robot is severely unbalanced (i.e., $\theta > 0$, $\dot{\theta} > 0$). In this case, the only possible way to balance the robot is to make $\ddot{\theta}$ as negative as possible (i.e., $\ddot{\theta} \ll 0$). To achieve this, we need to use a combination of large $F > (M + m)g\tan\theta$ and small $l$.
- The robot is inverted (i.e., $90° < \theta < 270°$ ($l < R_{\text{wheel}}$), $l = 0$, or $m = 0$). Now the robot is impossible to return to the desired balanced state (or the balanced state can't be defined), but becomes permanently stable.

In conclusion, a motor being able to produce huge $F$ is wanted for all the cases. Similarly, it's good to have small $m$, $M$, and/or $\frac{m}{M}$. Large $l$ is suitable for peaceful environment and high balance performance, while small $l$ or inverted robot is suitable for severe environment and high controllability.

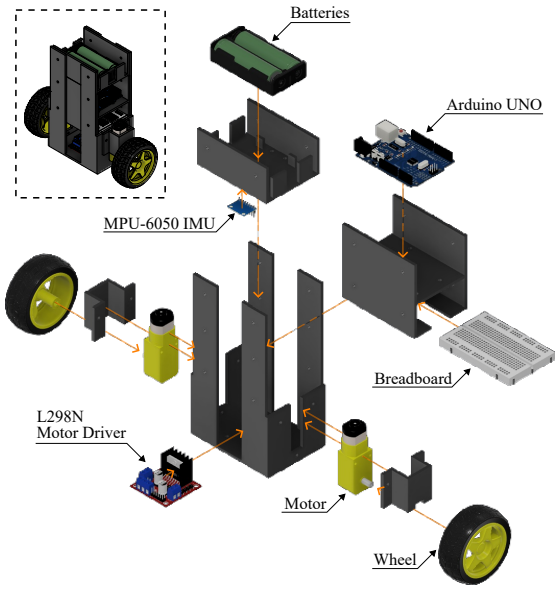
### 3.3.2. Chassis

We use CAD (Autodesk Fusion) to design the chassis model, and 3D-print the model. The chassis is designed with many components in the experimental stage to make some components easier to update. These components are assembled using M3 bolts and nuts. For the final design, all the components are combined into a whole one to make the chassis stable.

In Fusion, the chassis is designed with the help of the models of other electronic components. Official datasheets of the electronic components are also used. In this way, it's precise and accurate to design the size and position of chassis components and through holes. One of our CAD model designs is shown in Fig. 4. It has high mass center, which can perform good balancing skill but lack flexibility.




**(a)** Front-top-side view




**(b)** Robot components and assembly

**Figure 4.** A chassis model design.

PrusaSlicer is used to convert CAD models to 3D-printing code.

The printer uses PLA as the filament with diameter of $1.75mm$. To make our chassis sturdy, we use layer height of $0.3mm$. The vertical shells is set to 4 perimeters, and the horizontal shells is set to 4 top layers and 4 bottom layers. Since most of our chassis consist of plates of $3mm$, these parameters are sufficient to make our chassis solid. The fill density is set to 15%. The only disadvantage is that solid chassis may increase the mass. We compensated it by change the height of mass center by a larger value.

## 3.4. Signal Processing

### 3.4.1. IMU data processing

To get the robot state, we need to process the data from the IMU. The IMU contains a 3-axis gyroscope used to measure angular velocity and a 3-axis accelerometer used to measure acceleration. One important thing we can do is to get the angle (yaw, pitch, roll) from these values. Simply, we can integrate the 3 angular velocities $(\omega_x, \omega_y, \omega_z)$ to get the corresponding Euler angle. However, Euler angles suffer from Gimbal Lock, resulting in the loss of a degree of freedom and causing ambiguous calculations (for example, the results are the same whether it yaws by 30° and then pitches by 90°, or pitches by 90° and then rolls by 30°, which makes it difficult to determine the true yaw or roll angle). Because of this, we use quaternion $q = (q_0, q_1, q_2, q_3)$ to represent the angle state, which indicates the current state that is equivalent to rotate the initial state around an axis $(q_1, q_2, q_3)$ by an amount of $q_0$. The current quaternion is given by:

$$q_n = q_{n-1} + \frac{\Delta t}{2} \cdot q_{n-1} \otimes \omega \tag{5}$$

Another method to get quaternion is using the accelerations $(a_x, a_y, a_z)$. Because there is a constant gravity of $\approx 9.8m \cdot s^{-2}$, we can know the quaternion by comparing the direction of measured acceleration with the true gravity. However, this method is noisy, while using gyroscope can produce drift over a long time. We can mitigate the high-frequency noise of accelerometers by a low-pass filter, and the low-frequency drift of gyroscopes by a high-pass filter, which is called complementary filter.

There are many states we can calculate from the quaternion (Most calculations can be done by the IMU module or some libraries such as `i2cdevlib/Arduino/MPU6050`). For example:

$$\text{Pitch: } \theta = \arcsin\left(2(q_0 q_2 - q_3 q_1)\right) \tag{6}$$

$$\text{Yaw: } \psi = \arctan \frac{2(q_0 q_3 + q_1 q_2)}{1 - 2(q_2^2 + q_3^2)} \tag{7}$$

$$\text{Acceleration: } a_{\text{world}} = q \otimes (0, a_x, a_y, a_z) \otimes q^{-1} \tag{8}$$

In this way, we can get the tilt angles or accelerations much more precisely and unambiguously. However, to get the displacement of the robot, we need to integrate the acceleration twice, which can brings huge noise and drift. This disadvantage is mostly caused by the hardware, which is hard to improve by software. Therefore, we can choose to calculate the displacement from the signal used to control the speed of motors.

### 3.4.2. PID controller

PID (Proportional-Integral-Derivative) controller is the core of the feedback system. By comparing the current state $S(t)$ with the desired state, we get an error term $e(t)$. Using $e(t)$ as the input of PID controller, we can use the output $u(t)$ to drive the current state toward the desired state. The output expression is given by:

$$u(t) = K_p e(t) + K_i \int_0^t d\tau\, e(\tau) + K_d \frac{de(t)}{dt} \tag{9}$$

The Proportional term means that the larger the error is, the large the output is to restore the state. The Integral term means that the longer the error persists, the large the output is to further restore the state. The Derivative term means that the faster the error decreases, the smaller the output is to reduce oscillation. The three terms have individual coefficients $(K_p, K_i, K_d)$. Their combinations vary widely across different situations.

For the PID of balancing pitch angle, the 3 coefficients need to be tune along with the balanced pitch angle $\theta_0$. First, we found the approximate pitch angle. Based on this, we adjusted only $\theta_0$ and $K_p$ until the robots fell in both directions with a similar low probability, which means $\theta_0$ is closed to the true balanced pitch angle. Then, we tuned $K_d$ such that the robot wouldn't oscillate too much around the $\theta_0$. However, the robot may occasionally keep moving in one direction. We used large $K_i$ to correct it. The optimal tuning yields highly varied coefficients. For example, $K_p = 48$, $K_i = 600$, and $K_d = 0.19$ with PID sample time $= 2ms$. By tunning these coefficients and looking for a good balancing state, we can achieve better feedback performance.

### 3.4.3. Pulse width modulation

In order to control the robot smoothly and continuously, we need analog signals to do the task. However, the UNO has only digital output, which means the output voltage can be either HIGH level or LOW level without any values in between. PWM can solve this problem. Instead of encoding signals to voltage level, it produces a train of pulses as the output, and encodes signals to the width of each pulse. Meanwhile, our motor driver is able to receive the PWM signals.

However, the range of PWM values is limited. This disadvantage can be ignored in our cases, since the resolution of PWM is much better than the resolution of geared motors.

## 3.5. Control Logic

The program for controlling runs in Arduino UNO. The structure of control logic is based on 3.1 and Fig. 1, whose details are shown below:

### 3.5.1. Start

In this initialization sequence, all the components are configured. If anyone of them is not initialized successfully, the following controlling code would not run.

- `initImu()`: Sets the offsets of 3 axises for the gyroscope and the accelerometer, and waits until the IMU becomes stable.
- `initPID()`: Defines the balanced pitch and yaw, the PID coefficients for different situations, and the PID object for pitch and yaw.
- `initMotor()`: Uses `pinMode(pin, mode)` to initialize 2 EN pins and 4 IN pins.

### 3.5.2. Read States

In this process, we get the states of the robot from the calculations about quaternion. The logical code is shown in Code 1 For pitch and yaw angles, we calculate the error between the true angle and the balanced angle, unwrap the $2\pi$ period, and use the result as the input of feedback control. For accelerations, we compute the value in the world frame using the quaternion.

```
double unwrapAngle(float angle, double balanced) {
  double a = angle * 180 / M_PI - balanced;
  // ... unwrap from 2*pi ...
}
bool getRobotState() {
  imu.dmpGetQuaternion(&quaternion, imuFifoBuffer);
  imu.dmpGetYawPitchRoll(imuYpr, &quaternion, &gravity);
  pitchAngle = unwrapAngle(imuYpr[1], balancedPitch);
  // ... doing similar jobs for yaw and acceleration ...
  imu.dmpGetLinearAccelInWorld(...);
```

```
}
```

**Code 1.** Example code of "Read States" process

#### 3.5.3. *Check States*

The states gotten from the previous process is analyzed, and the next step is determined. For example (as shown in Code 2), the coefficients of PID controller will dynamically transition to surviving mode (high-gain feedback) if the pitch angle is too large, or the motors will be disabled if the robot is destined to fall.

```
if (abs(pitchAngle) > maxPitchAngle) { motorStop(); }
// ...
if (abs(pitchAngle) > survivingPitch) {
  pitchPID.SetTunings(sKp, sKi, sKd);
} else { pitchPID.SetTunings(kp, ki, kd); }
```

**Code 2.** Example code of "Check States" process

#### 3.5.4. *Feedback Control*

Our feedback control is currently based on only PID controller. We use a library to manage the computing. In the code, the feedback output can be gotten from `pid.Compute()`. The code is simple, while this process is essential to the robot. All the coefficients need plenty of time to tune manually.

#### 3.5.5. *Take Action*

This process convert the digital signals from feedback control into a real action of the robots. For example, accelerating immediately, or rotating clockwise slowly. The action is also influenced by "Check States". As shown in Code 3, balancing the pitch is a priority over adjusting the yaw.

```
if (abs(motorValue) > maxMotorValue * 0.3) {
  motorMove(motorValue, ENA_PIN, IN1_PIN, IN2_PIN);
  motorMove(motorValue, ENB_PIN, IN3_PIN, IN4_PIN);
} else {
  motorMove(motorValue + yawCorrection, ENA_PIN, ...);
  motorMove(motorValue - yawCorrection, ENB_PIN, ...);
}
```

**Code 3.** Example code of "Take Action" process

## 4. PROTOTYPE PROGRESS

### 4.1. Current Prototype Capabilities

During the last 6 weeks, we have finished these tasks: connecting and initializing each components, reading angles and accelerations from IMU, using PWM to driver motors, balancing the robot by its pitch angle with PID, controlling the yaw angle basically.

The robot of current version is shown in Fig. 5.

Preliminary performance metrics of the current prototype are listed in Table 4. Most top-level requirements mentioned in 2.2 are achieved. Some evaluations are even better than expected, such as the max displacement in 1s without perturbation. However, the max allowed perturbation pitch ($< 15°$) is not achieved. It's because the high mass center can weaken its perturbation resistance. To improve this skills, we will try to increase the strength of motor driver, increase the wheel size, or decease the mass and the height of mass center.

### 4.2. Current State

It's the 6th week. We are mainly focusing on the reports and the review of the last 6 weeks. Meanwhile, we also start to design or improve the prototype to achieve the advanced requirements. These tasks are being done: controlling the yaw angle with high flexibility, and controlling the linear motion at a constant speed with good balance performance.

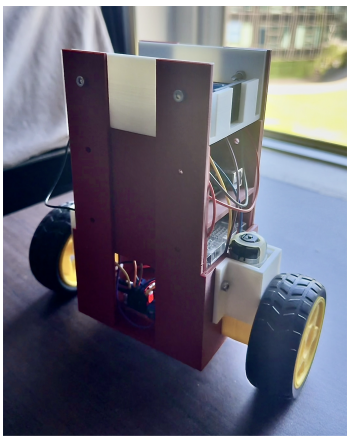


**Figure 5.** The robot with complete basic balancing skills and preliminary advanced skills. It can stably balance itself on a small table.

| Evaluation Metric | Rough Value |
|---|---|
| Balancing Time (no perturbation) | $\sim \infty$ |
| Max Pitch (no perturbation) | $\approx 3.5°$ |
| Max Displacement in $1s$ (no perturbation) | $\approx 0.15\ m$ |
| Min Pitch to Drive Motors | $< 0.4°$ |
| Perturbation Reaction Time | $\ll 0.2\ s$ |
| Max Allowed Perturbation Pitch | $\approx 13°$ |
| Max Time to Restore from Pitch Perturbation | $\approx 1.5\ s$ |
| Max Displacement by Pitch Perturbation | $\approx 1\ m$ |
| Max Time to Restore from Yaw Perturbation of $180°$ | $\approx 3\ s$ |

**Table 4.** Some key evaluations of the robot of current version.

## 5. PROJECT MANAGEMENT

### 5.1. Responsibilities

Initially, the project was divided into two parts: hardware and software. The origin responsibilities are listed below:

- Lin: hardware part, including electronic components and mechanical design
- Yang: software part, including signal processing and control logic

However, our progress was unexpectedly fast. The first version of balancing robot was developed much earlier than planned. After discussions, we both agreed that designing the hardware and software together is much more efficient than designing them individually and then combining them. By team working and peer learning, we quickly mastered both the hardware and software part. Therefore, we use a semi-parallel structure for our project. Both of us is responsible to design a robot with different skills individually while sharing the same mechanical and logical architecture. There are also regular discussions about integrating these skills into one robot. Our current responsibilities about skill designing are listed below:

- Lin:
  - Locking skill: the robot can lock itself to a certain direction, even if it's rotated by us deliberately.
  - Adaption skill: the robot can judge the situation that it's facing or will face.
- Yang:
  - Motion skill: the robot can move forward and backward actively while keeping balanced.

– Resurrection skill: the robot can restore itself from a heavy perturbation.

Combining these skills, we can achieve the advanced requirements mentioned in 2.2

### 5.2. Estimated Timeline

The remaining development will be divided into 3 phases:

- Phase 1: Improve the locking skill and motion skill. Combine them into one robot.
- Phase 2: Develop the adaption skill and resurrection skill. Discuss the development plans in the beginning, and combine the development results at the end.
- Phase 3: Conduct comprehensive tests. Do the final improvements based on the performance.

The detailed timeline is shown in Fig. 6

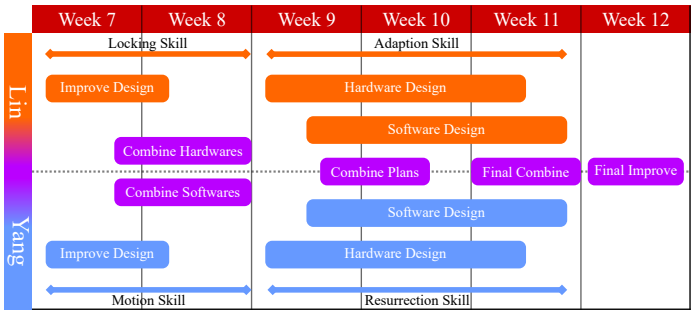



**Figure 6.** The Gantt chart of our estimated timeline.

### 5.3. Budget

The project was designed to be cost-efficiency for rapid prototyping, utilizing off-the-shelf hardware and open-source software libraries. While the primary components were provided for this study, the total retail cost for the electronics and mechanical parts is approximately $120 AUD. Additionally, iterating the chassis design via 3D printing cost less than $5 AUD in material costs. This low-cost approach ensures the platform remains accessible for replication and future development.

## 6. CONCLUSION

This design review has detailed system architecture, mathematical and physical foundations, and developmental progress of an autonomous balancing robot. By leveraging a Lagrangian dynamic model, the physical chassis was currently optimized with an elevated center of mass. System requirements have been addressed through a robust electronics stack, utilizing an $I^2C$ MPU-6050 IMU. A 500 Hz PID control loop effectively translates error calculations into physical restoration via the L298N motor drivers. Current prototype testing confirms the viability of this design; the robot successfully maintains steady-state dynamic equilibrium with minimal drift and recovers from impulse perturbations up to 13°.

While the current perturbation resistance falls marginally short of the 15° target, the identified hardware limitations provide a clear pathway for refinement. As the project enters the final phases of development, the transition to a concurrent engineering strategy ensures that advanced mobility and adaptation skills will be integrated efficiently.